

RESEARCH STATEMENT

KEATON QUINN

1. OVERVIEW

My research interests lie in differential geometry. Specifically, my research projects often seek to answer geometric questions using analytic tools, typically from linear algebra and differential equations. These questions usually involve Riemannian manifolds, which are topological spaces modeled on Euclidean space and equipped with a notion of distance and angle based on an inner product attached to each tangent space to the manifold. These inner products bundled together are called a Riemannian metric, and this metric lets us investigate the geometry of these spaces. For example, the notion of a straight line in this setting is captured by geodesics, and these are locally solutions to a system of nonlinear ODEs with coefficients defined from the metric. The curvature of a Riemannian manifold may be thought of as the obstruction to the manifold being locally isometric to flat Euclidean space, and it can be expressed solely in terms of the metric and its derivatives. My work frequently deals with similar topics in the setting of hyperbolic manifolds: manifolds with curvature everywhere equal to -1 , meaning they are saddle-shaped at every point and in every direction.

My main research techniques deal with using infinitesimal data on the boundary at infinity of hyperbolic space (in the form of tensor fields) to construct surfaces embedded in its interior. The main example of this idea is the construction due to Epstein in [Eps84]. In [Qui20] I used these Epstein surfaces to identify certain families of surfaces in hyperbolic manifolds that have nice behavior as they head off to infinity (meaning as they leave the ends of the manifolds). The constant (Gaussian) curvature surfaces of Labourie form one of these nice families, so my results were able to resolve a conjecture of his from [Lab92]. The constant (mean) curvature surfaces of Mazzeo and Pacard [MP11] also form one of these nice families, so my results dictate their asymptotics as well.

This construction of Epstein is actually a duality and exists in any dimension. Indeed, the geometry of hypersurfaces in hyperbolic space is governed by the Gauss-Codazzi equations, and it has been shown [KS08] that there is a dual set of equations governing the corresponding infinitesimal data at infinity. In dimension 2, all solutions to these dual equations were identified by [BB22]. In my own work [Qui23], I have identified the form these dual equations take in higher dimensions, established similar existence and uniqueness results for their solutions, and related the solutions to well-known tensors that control the conformal properties of the data at infinity.

More recently, I have enjoyed finding ways of applying my geometric interests to other settings as well as incorporating undergraduates in my projects. I am a mentor for the undergraduate fellows program and for the high school summer research program at Wellesley College's Institute for Math and Democracy. I have led projects on investigating the utility of replacing single member political districts with multi-winner districts using the Markov-chain methods developed by [Dat25], [DDS21]. Another project consisted of using geometric methods to measure gerrymandering in states that have redistricted after 2020. Future projects related to this will investigate how considering the curvature of the boundary of a district might improve these geometric methods. I am also working on expository differential geometry projects that are accessible to undergraduates familiar with linear algebra. I intend to enlist the help of students to investigate these topics in higher dimensions. The following sections expand upon these recent activities and describe the directions I intend to take them as undergraduate research.

2. FUTURE DIRECTIONS AND STUDENT PROJECTS

This section details future project ideas, both for myself and as undergraduate research. The first few relate to the social choice theory mentioned above. The remaining are an expository geometry investigation as well as an application of mathematics to student interests in video games. I have highlighted the main questions or conjectures related to each project. I also indicate where undergraduate involvement is possible, and what mathematical background is suggested.

Measuring Gerrymandering Using Geometry. Many states require their political districts to be “compact” as a way to prevent bad actors from intentionally drawing political maps to favor one party over others. The idea is that the less compact a district is the more likely it was drawn with bad intentions. But the definitions of compactness required by states are usually not well-defined, in addition to not being that of the mathematician. Several ways of measuring compactness have been proposed in the form of “geometric compactness scores”. These numerical scores try to indicate how round or convex a district is, where seemingly the more round a district is the better. One such compactness score is based on the classical isoperimetric inequality, which says that the enclosed area A and perimeter P of a curve in the plane satisfy $4\pi A/P^2 \leq 1$. Equality is attained only when the curve is a circle, so the value of the ratio $4\pi A/P^2$ is some indication of how circular the curve is. When the curve is the boundary of a political district, this ratio is called the Polsby Popper score [PP91].

During Summer 2025, I was a project supervisor for the high school summer research program at Wellesley’s Institute for Math and Democracy. We investigated the change in various compactness measures, including the Polsby Popper score, before and after a state redrew their political district maps. This change in scores is one measure of potential gerrymandering, since individual compactness scores on their own can not say much. We then performed a statistical analysis based on a sample of states that have redrawn their maps since 2020 and concluded that, on average, there is no change in the mean compactness score of a state after redistricting.

Because it is possible to deform a curve by drastically increasing its perimeter while leaving its enclosed area roughly the same, the Polsby Popper score is not very robust. For example, districts that follow rivers, coastlines, or political boundaries can have poor Polsby Popper scores due to long perimeters, while appearing to the human eye as compact. Recently [S⁺25], others have investigated using ideas from differential geometry to measure compactness, in this case using the curve shortening flow. This is a geometric flow (essentially a PDE) that deforms a curve according to its curvature. The result is that over time, the curve’s length decreases and the enclosed region becomes circular. The authors of [S⁺25] use the amount of time it takes a specific district to become circular as the measure of that district’s compactness.

I intend to lead undergraduate projects investigating how hitting every district in a state with the same short-time curve shortening flow changes Polsby Popper scores. A short-time flow would smooths out and decrease the perimeters of districts giving Polsby Popper scores closer to what our eye would expect. We would then re-perform the statistical analysis to see how much of our previous work depended on immutable characteristics of the state, like a district following a coastline.

Question. *Let $t > 0$. For each boundary of each district in a state, apply the time- t curve shortening flow. How does the statistical analysis of Polsby-Popper scores depend on t ?*

I believe this is accessible to those familiar with multivariable calculus and linear algebra, and will provide students the opportunity to meet differential geometry, which is not usually a requirement for graduation.

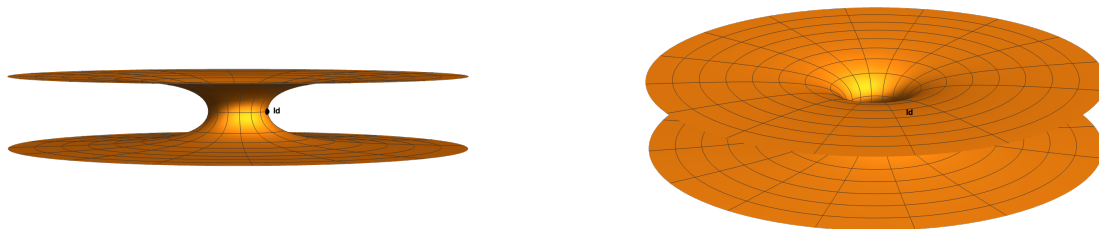
More generally, I intend to investigate ways of drawing political district maps using variations of the curve shortening flow. Here we would apply the geometric flow to the whole map at once.

Boundaries between districts would compete in shortening and individual districts should become more round. This could provide ways to draw district maps for states that avoid partisan bias.

Multi-member Districting. States in the US elect a single candidate per district to the House of Representatives. Depending on how district boundaries are drawn, this may reduce representation of minority voters by splitting them across majority districts. Among many potential preventions of this scenario is the idea of using multi-member districts, where more than one candidate would be elected per district. This potential was investigated in [DP⁺22] on the congressional level where it was determined that using multi-member districts with ranked choice voting leads to more demographically accurate representation of the voters in the districts. This past academic year I supervised an undergraduate project that began, but did not finish, investigating these topics on the state legislative level. Our goal was to reproduce the Markov-chain methods used by [DP⁺22] on the state of Pennsylvania. I intend to pursue this idea further with students in the future. In my experience, they are naturally interested in these questions, and being able to focus these computations on their home states gets them invested in the math behind it. These ideas are approachable to students with exposure to multivariable calculus, and familiarity with programming.

Expository Geometry. I have been investigating standard differential-geometric quantities for low dimensional manifolds with the intention of producing geometrically accurate images of these objects that naturally live in $\mathbb{R}^4$ or higher. One such manifold is $\text{SL}_2\mathbb{R}$, the set of 2×2 matrices with determinant 1. This unbounded set has dimension 3, being the level set of a smooth function (the determinant) on the 4 dimensional vector space of all 2×2 matrices. Hence it is hard to picture in $\mathbb{R}^3$. Much is known about $\text{SL}_2\mathbb{R}$ with respect to the Riemannian metric given by $\langle M, N \rangle = \text{tr}((A^{-1}M)^t A^{-1}N)$ on the tangent space at A . In many ways, this is the natural metric to consider as it is invariant under left multiplication. In other words, under this metric, the group $\text{SL}_2\mathbb{R}$ acts on itself by isometries. There is a rich theory here that is well known.

However, this is not the natural metric on $\text{SL}_2\mathbb{R}$ when it is considered as a subset of 4 dimensional Euclidean space. If we could view 4 dimensions, we would instead see $\text{SL}_2\mathbb{R}$ under the Euclidean metric given by $\langle M, N \rangle = \text{tr}(M^t N)$. It is with respect to this metric that I have been computing curvatures and finding geodesics. Starting at the identity, there are three directions distinguished by the geometry. If we follow the geodesic in the direction of one of them, call it v , we return to the identity. This circle that has been traced out is in fact $SO(2)$, the rotation matrices. If we consider the set of matrices produced by following v and one (actually any) of the other directions, we obtain a 2 dimensional surface in $\text{SL}_2\mathbb{R}$. This surface is topologically a cylinder, and I have found a cylinder in $\mathbb{R}^3$ that has the exact same curvature. By considering this geometrically accurate 2-dimensional “slice” of $\text{SL}_2\mathbb{R}$ we get a glimpse into what this 3 dimensional shape looks like in $\mathbb{R}^4$.



Some of this has been reproduced in higher dimensions, but in general the $n \geq 3$ case is slower. For example, the formula for the inverse of a 3×3 matrix is more complicated than the nice 2×2 case, and this makes it more challenging to find explicit solutions to these geometry calculations.

Questions.

- Can we find explicit formulas for integral curves of the gradient of the determinant function on $GL_n\mathbb{R}$ for any $n \geq 3$?
- Can we find explicit closed formula expressions for geodesics in $SL_n\mathbb{R}$ starting at the identity?
- Can we express $SL_n\mathbb{R}$ as the product of the $n^2 - 1$ geodesics that emanate from the identity in the same way we can when $n = 2$?

Many of these ideas reduce down to matrix computations. This means this setting is approachable to undergraduates familiar with linear algebra, and I would like to enlist them to see what can be done for these higher dimensions.

Classifying Video Games. The video game industry is now the largest grossing entertainment industry in the world [IG24], surpassing the film and music sectors combined. Indeed, 85% of teenagers in the united states (97% of boys and 70% of girls) report playing video games [GS24]. In fact, not only are most of our students playing games, but they may be becoming better geometers from doing so [MGBPI23], [FSP07]. One of my goals as a mathematician is to show our students how math can be applied to their interests, and to this end I have several projects in mind related to gaming.

Certain games in the action-adventure genre focus on exploration. Many times, player progress is blocked by obstacles that require power-ups, and entertainment value is gained from searching for these. For example, the entrance to a new area may require the player to jump higher than they are currently able. Once they obtain this ability, they can return to the area and make progress towards beating the game. To simplify the discussion, call these barriers to progress *locks* and the abilities needed to pass them *keys*. So in the example, the lock is the entrance being too high, and the key is the high jump ability. By considering these locks and keys, a video game can be encoded using binary decision diagrams and graph theory, and this is an emerging area of research [MRM22].

Usually the keys need to be collected in a certain sequence in order to complete the game. However, many of these games have modifications, frequently coded by their fans, called *randomizers*. See [h⁺25], [t⁺25], or [DbsN⁺25] for popular examples. Here the in-game locations of the keys are permuted, and this adds variability (hence novelty and excitement) to new play-throughs [Whi19]. One issue, however, is that keys may be placed behind the locks that require them to open, making the game not winnable. Typically these cases are excluded by the code governing the randomizer. But this leads to following:

Question. *Given a video game, what percentage of its randomizers are completable?*

This is a great way to get an undergraduate interested in mathematics. We can ask this question about their favorite video game (if they have one) and learn the theory behind its solution.

But we can go a bit further. In differential geometry (and other fields) we often consider moduli spaces of objects to help us study classification problems, or to help make intuitive notions precise. In analogy to this, we could consider the ‘moduli space of randomizers’ of a video game. The size of the moduli space can then be thought of as some measure of how complex the video game is. Moreover, the size of the portion of winnable games in the moduli space can be thought of as measuring how linear a game is, where we call a game linear if the locks must be found in exactly one order. For example, if the size of the set of randomizers is large but the size of winnable randomizers is smaller, we could imagine the game is complicated, but relatively linear, since only a few permutations of the keys keep the game completable.

This project would introduce an undergraduate researcher to these math ideas. For example,

Question. *Is there appropriate equivalence relation to place on the space of randomizers of a video game to form a quotient moduli space?*

We would try to make these notions precise and compute examples for several popular games in this genre. We can make an ordered list of games according to their linearity. And one could imagine using this to produce recommendations for games to play similar to the ones a person likes. This might also be a useful to game designers as a tool for identifying if there are accidental hard-locks in their games, i.e., unintended situations that prevent players from completing the game.

A variation of item randomization is the idea of a *room randomizer*. Here the transitions between rooms in the game are permuted, but usually in a way that preserves transition orientation. Some relevant questions here are

Questions. *Given a video game, is the graph that records the randomized room transitions planar? Are all such graphs for the video game planar?*

The relevance of this is the following use. If they are planar, that means an easily readable map of the game respecting the new room transitions could be presented to the player, and viewable on their 2-dimensional screen.

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